INCH-POUND
MIL-DTL-24643/40E
22 August 2002
SUPERSEDING
MIL-C-24643/40D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SMU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, tin coated, ASTM B 286, AWG Size 22, 7 strands.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth -	Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent.
Fifth -	Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.
Sixth -	The required number of singles (see table I) cabled around a center filler with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1. Fillers shall be used as necessary to form a firm well rounded cable.

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Construction (non-watertight) (continued)

Seventh - Binder tape, applied helically with overlap.
Eighth - Cross-linked polyolefin jacket. Minimum average thickness 0.063 inch.

TABLE I. Details.

Military part no. M24643/40	Size	Number of Conductors	Diameter over completed conductor (inch) (nominal)	Overall diameter (inches) TYPE LS1SMU	
				min	max
-01UN	5	5	0.131	0.465	0.500

EXAMINATION AND TESTS:

Basic electrical: Requirements

Conductor resistance - ohms/1000 feet at 25°C, maximum 17.37
Voltage withstand - volts, root mean square, minimum
Conductor to shield 5000
Shield to shield 500

EXAMINATION AND TESTS (continued):

Requirements

Insulation resistance - Megohms -1000 feet minimum
Conductor to shield 500
Component shield to shield 100
Conductor and shield continuity ..... No failure

Group A:

Visual and dimensional ..... No failure
Characteristic impedance - at 1MHz, ohms ..... 50 ± 2

Group B:

Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C) 50
Attenuation - at 400 MHz, decibel (dB)/100 FT maximum 18
Cold bending, cable - (see table I for mandrel diameter) ..... No damage
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250

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Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals - (aged)
Insulation
Air Oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air Oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45

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Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-038)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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